

Q&A Preparation Notes

For *CALM-Net_final_presentation.pptx*. Every number comes from the results files or the manuscript. Regenerate with *presentation/qa_notes.py* whenever the results change.

1. Datasets and trials

	A, exoskeleton	B, treadmill	C, motor exec.	D, BCI IV- 2a	E, exoskeleton
Source	OpenNeuro ds007788	MoBI (Luu 2017)	PhysioNet EEGMMIDB	BNCI2014-001	DECODED / EUROBENCH
Participants	7	8	20	9	7
Recordings	9 sessions over weeks	3 trials	6 motor-exec. runs	2 sessions, different days	runs within a session
EEG	60 ch, 100 Hz	64 ch, 100 Hz	64 ch, 160 to 100 Hz	22 ch, 250 to 100 Hz	27 ch, 200 to 100 Hz
Task	walk vs stop	walk vs stand	movement vs rest	left vs right hand	walk (MI) vs stand
Fitted on	sessions 1-3	trial 1	runs 3, 5, 7, 9	session 1	first 8 runs
Tested on	sessions 4-9	trials 2-3	runs 11, 13	session 2	remaining runs
Window / stride	4 s / 0.5 s	2 s	2 s / 0.5 s	4 s, one per trial	4 s / 0.5 s
Class blocks	789	66	3600	734	192
Median block	18 windows	214 windows	5 windows	1 window	13 windows
Longest block	486	2211	5	6	33

- Cohort A in words: about 5031 fitting windows per participant.
- Cohort D in words: 288 trials per participant, but only about 100 fitting windows, the fewest of the five cohorts. That is why five of the eight published decoders are left near chance there under shared hyperparameters.
- Cohort E in words: each run is 15 s standing, 24 s walking under kinesthetic motor imagery, 22 s walking under a counting distractor, 14 s standing. We decode standing against walking under motor imagery and drop the counting condition rather than folding it into either class.
- If asked why the cohort B block length changed from earlier drafts: it is 214 windows, measured on the fitting recordings over a contiguous timeline. The earlier figure of 309 was measured on post-split subsets, which are not contiguous, so it was wrong.

2. Model architecture

- 290,943 parameters. The input is a 60-channel, 4-second window on cohort A.
- There are two paths and nothing else.
- 1. Multi-scale power stem. Three temporal resolutions (64, 128, 256 ms), each with depthwise spatial filters, then square, pool, logarithm. This is log band power per channel, which is the diagonal of the window's spatial covariance. The design follows ShallowFBCSPNet.
- 2. Tangent-space branch. Form the window covariance C , shrink it toward a scaled identity at $\lambda = 0.1$, and map it to the tangent space at a reference M : $\phi(X) = \text{vec}(\log(M^{-1/2} C M^{-1/2}))$, with off-diagonal entries scaled by root 2 so the Euclidean norm equals the matrix

Frobenius norm. 1830 dimensions for 60 channels, projected to 128. This reads what the stem discards.

- The two 128-dimensional outputs are concatenated and fused into one linear classifier.
- The reference M is a running mean over incoming window covariances. It uses no labels and keeps updating at inference.

Component	Parameters
Multi-scale power stem path	19,505
Tangent-space branch	238,028
Fusion layer	32,896
Output layer	514
Total	290,943

- If asked why the tangent map carries no gradient: backward through an eigendecomposition contains $1/(\lambda_i - \lambda_j)$ terms, which diverge on near-degenerate spectra, and covariances from 60 channels and 400 samples are routinely near-degenerate. The projection after it is learned.

3. Contributions

- 1. A tangent-space branch beside the log-power stem, with a reference estimated from the input alone and no labels used at inference.
- 2. Measured on five cohorts and 51 participants, three seeds each. Accuracy rises on four: cohort C 0.779 to 0.843, which is +0.064 ($p = 8.8e-5$, 20 of 20); cohort A 0.864 to 0.909; cohort D 0.701 to 0.741 ($p = 0.020$); cohort E 0.928 to 0.949 ($p = 0.047$). It costs 0.175 on cohort B, 0.744 down to 0.568.
- 3. Condition 1, a two-sided rate condition: the estimator memory $\mu = N/m$ must be longer than a class block and shorter than the drift timescale. Read from the protocol alone, it separates all five cohorts with nothing fitted.
- 4. A causal test of that condition, by moving the memory directly on one cohort and switching the effect off and on in both directions.
- 5. A comparison with eight published decoders trained in one pipeline with shared splits, seeds and metric code, including the cohort where most of them fail to train.
- 6. A stronger version of the condition, pre-registered and reported as falsified.

4. Limitations, say these before you are asked

- The block-length boundary was found after the fact. It separates all five cohorts with nothing fitted, but it was identified after cohorts A, B and C had run. Only its application to D and E was prospective.
- The artefact control rests on seven participants. ICA cleaning takes accuracy from 0.897 to 0.859, a change of -0.038 ($p = 0.58$, four of seven unchanged or better). The mean is carried by sub-06 alone at -0.207, and without it the change is -0.010. It constrains the artefact account; it does not exclude it.
- Small cohorts: 7, 8, 20, 9, 7. Between-subject SD of 0.06 to 0.13 is large relative to the differences between arms.

- The branch is not compact. 238,028 parameters against a 19,505-parameter stem path, so we make no parameter-efficiency claim.
- Offline only. No closed-loop testing on a real exoskeleton.
- The upper bound of Condition 1 is not measured. Our recordings bound the drift timescale only from above, at about one session.
- Cohort E tests task, not drift: only one participant recorded on two dates, so its split is temporal within a session. It is the closest match to the target application and the weakest test of longitudinal shift.
- Best-epoch model selection is upward-biased on small folds. It applies equally to every arm and baseline in a pipeline.

5. Why this had not been done

- Data: long multi-session exoskeleton EEG was rare. ds007788 gives 9 sessions over weeks per participant; earlier exoskeleton datasets were single-session.
- Separate fields: Riemannian alignment estimates its reference over a complete recording, offline, so the recording must exist first. Test-time adaptation removes that constraint but was developed for images. We put the two together inside one network layer.
- Hidden failure mode: the rate problem only appears on protocols whose block lengths differ by orders of magnitude. Ours span 1 to 214 windows. With a single dataset you would never see it.
- Say 'to our knowledge'. Never say 'nobody has done this'.

6. Technical questions

- Validation split. 30 per cent of the class blocks from the fitting data, chosen per class, with whole blocks kept together so overlapping windows cannot leak. Used for early stopping, best-epoch selection and temperature scaling. Test recordings are never used for training or selection. The branch sees test EEG but never test labels.
- Training. AdamW, lr $3e-4$, weight decay $1e-2$, one-cycle schedule. Batch 32, at most 30 epochs, early stopping after 12 without improvement, gradient clipping at 1.0, class-weighted cross-entropy. Hardware: RTX 2060 and RTX 5080, PyTorch with braindecode and MNE-Python. Do not quote a per-model training time; there is no clean measurement.
- Seed variance. Three data-split seeds. A seed changes the validation blocks, the weight initialisation and the batch order. Across the five cohorts the reported model spans 0.007 to 0.022 and the stem spans 0.011 to 0.057, so the stem is the noisier arm on all five. We do not interpret differences below 0.02, and differences below the relevant seed spread are reported without a ranking.
- Statistics. Paired Wilcoxon signed-rank over participants, seed-averaged within participant, with Holm correction where there are multiple comparisons.
- Metrics. Balanced accuracy, because classes are imbalanced. Expected calibration error with 10 bins. Spurious activations per minute of standing. Missed onsets. Detection latency.
- Preprocessing. 8-30 Hz zero-phase band-pass, which excludes the 1-2 Hz gait rhythm and most muscle power above 30 Hz. EOG regression on cohort A, none on B to E. Amplitude

normalisation globally per recording, before the squaring stage. No ICA in the main results; the ICA control is reported separately.

- Adaptation rate. Momentum $m = 0.2$ with batch $N = 32$ gives $\mu = N/m = 160$ windows. Cohorts A (18), C (5), D (1) and E (13) have blocks shorter than 160, so the branch is admissible. Cohort B (214) does not.

7. The questions that will actually be asked

- 'Isn't cohort C just your best result?' No, and it is the one cohort where that cannot be true. Every design choice was fixed on A and B before C was run. That is what held out means here, and it is why C carries the accuracy claim rather than A, where $n = 7$ and $p = 0.22$.
- 'You failed on cohort B.' Yes, by 0.175, and we predicted it before training from the protocol. Cohort B presents walk in blocks of 214 windows against an estimator memory of 160, so the reference converges onto the walking covariance and whitens away the difference. Enlarging the memory recovers 74 per cent of the loss.
- 'Are you decoding muscle, not brain?' Three things say probably not, none of them conclusive. ICA cleaning costs 0.038 and is not significant. The loss does not follow how much muscle was removed ($\rho = +0.05$, where dependence would be negative). And band power falls over sensorimotor cortex during walking, which is the sign of μ and β desynchronisation; gross artefact would raise it.
- 'Why is the branch worth 238,028 parameters?' On the held-out cohort it is worth +0.064 and it beats the best published decoder by 0.060. We make no efficiency claim, and the parameter count is in the architecture figure.
- 'Accuracy went up but is the controller better?' Not everywhere, and we report that. Calibration improves on four of five cohorts and missed onsets fall on four, but on cohort A spurious walk commands rise from 0.70 to 2.09 per minute of standing. On a device that moves someone's legs that is a real cost.
- 'Is fusion doing anything, or is it just the branch?' Class separation measured in the trained model's own activations is 0.74 for the stem, 1.22 for the branch and 2.41 after fusion. Higher than either path alone, so they are complementary.
- If you do not know a number, say you will check it. Do not estimate one aloud.